How data science changed between 2017 and 2021

Eight trends from five years of the Kaggle Data Science Survey, and what they imply for planning

THE BIG IDEA

In five years data science became a **younger, Python-first, globally sourced** profession. The skills and tools settled quickly. What did not move is the share of employers running models in production: **stuck at about one in three** since 2018.

What this deck does

Eight slides of evidence, one trend each, then a recap and three implications for any organisation planning its data science hiring, tooling and training.

Five annual surveys, 106,301 practitioner voices: the largest open record of how the field works

106,301

responses, 2017 to 2021

16.7k to 26.0k

respondents per year, up 55%

293

questions on roles, tools, methods, learning

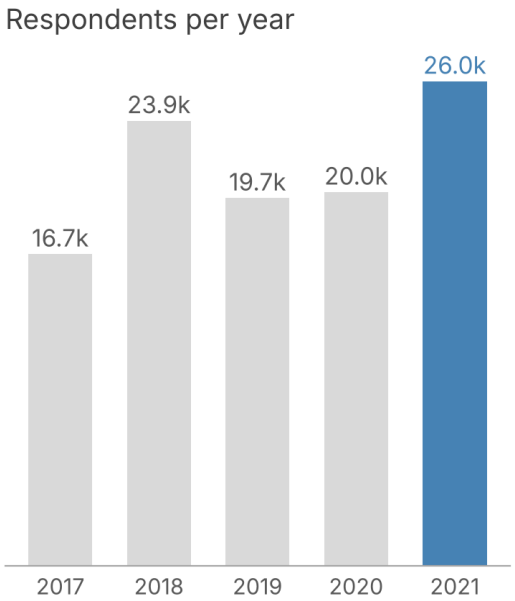
How to read the numbers that follow

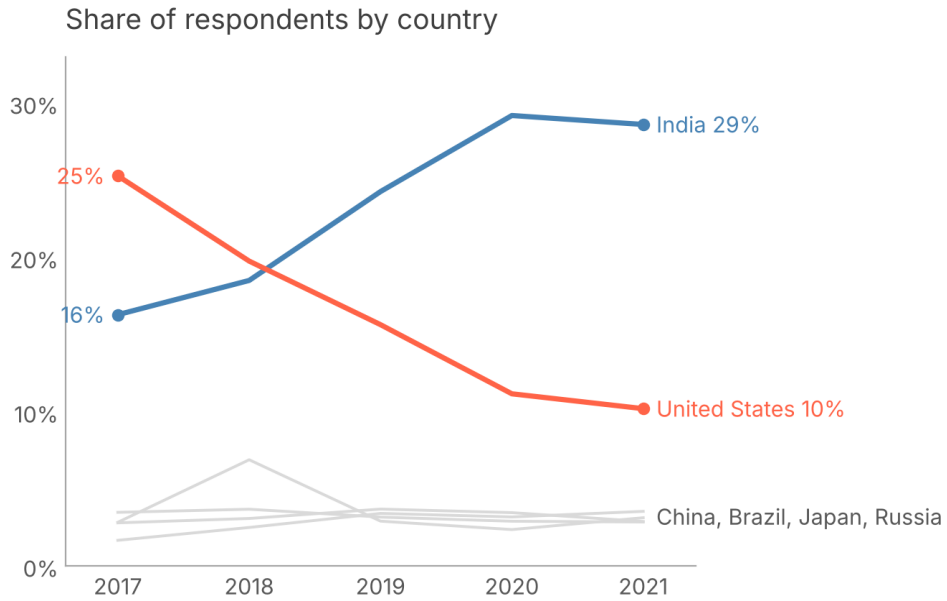
Every percentage is the share of respondents who answered that question in that year. Kaggle reworded answers most years, so categories were matched first (Female and Woman, Scikit-Learn and Scikit-learn, both spellings of China).

It is a self-selected sample of Kaggle users, not a census of the profession. Read what follows as directions of travel, not exact market shares.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

The community grew 55% in five years, and its centre of gravity moved from the United States to India



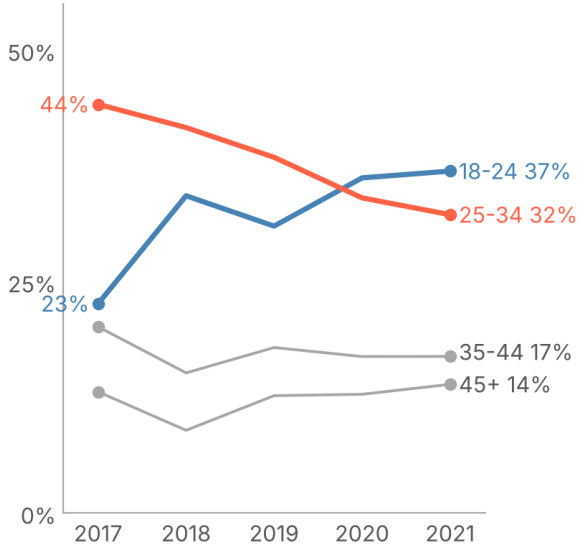


India overtook the US in 2019 and now supplies **nearly 3 in 10 respondents**; no other country exceeds 4%. The survey increasingly describes a young, global community rather than a US-centred one.

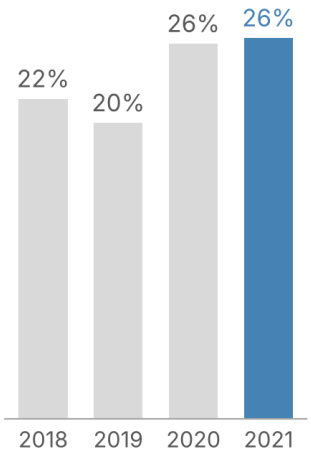
Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

The community is getting younger and more student-heavy, but no more gender-balanced

Share of respondents by age group

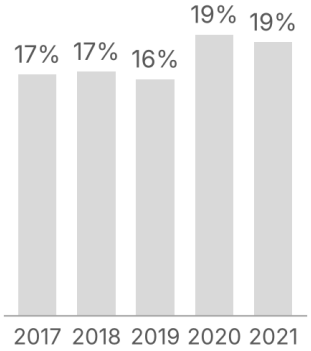


Share who are students



Share who are women

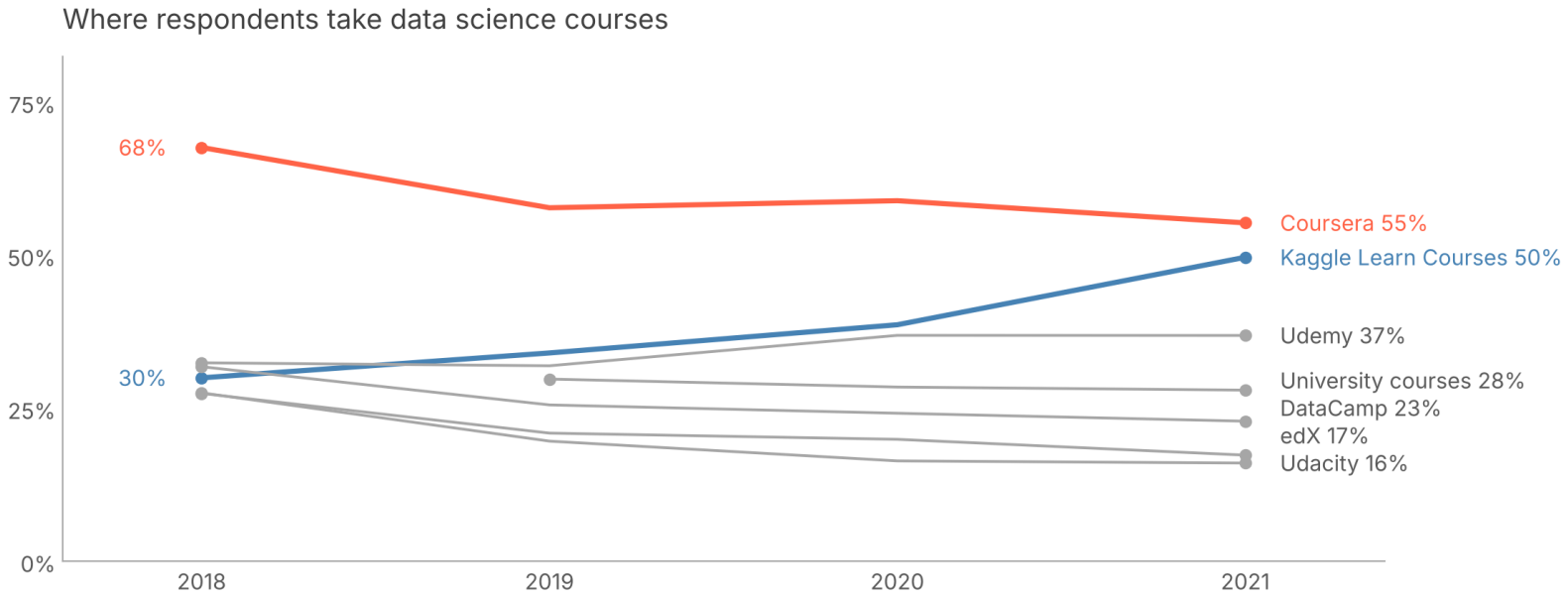
never above one in five



The 18 to 24 group grew from 23% to 37% and is now the largest; students are a quarter of respondents. **Women stay under one in five** every single year.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

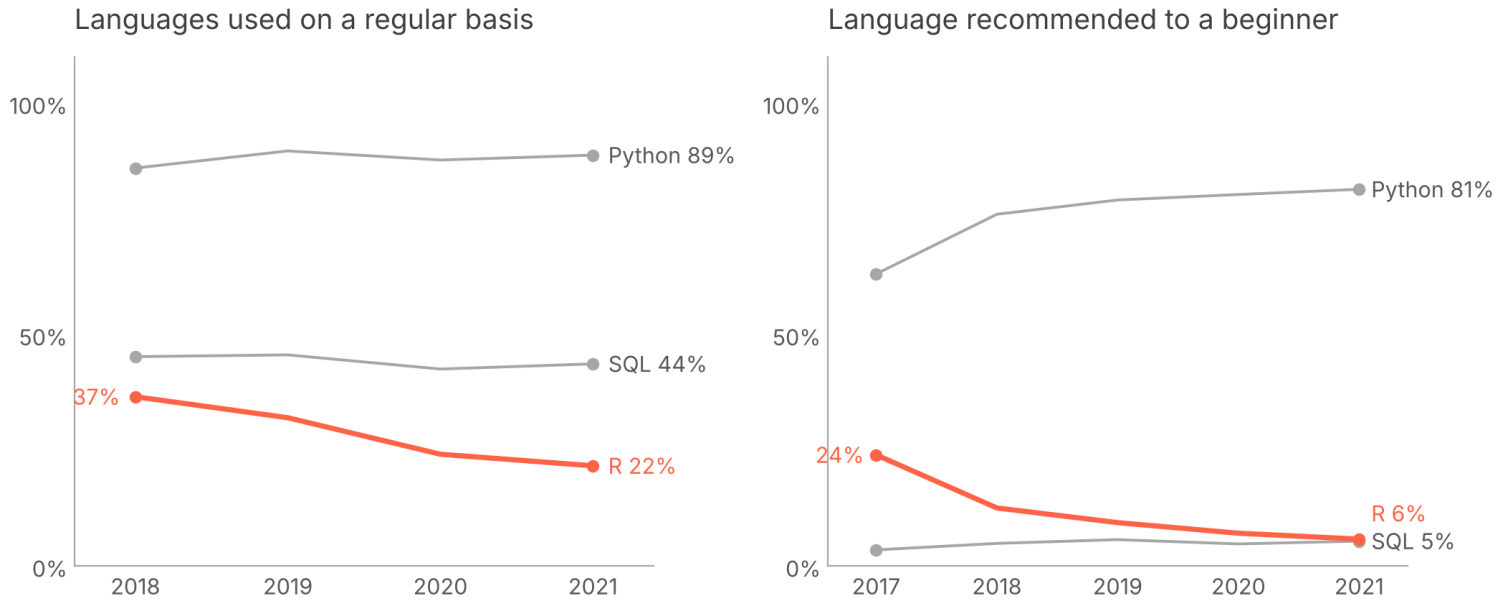
Learning has moved to free, hands-on platforms; the big course providers are all losing share



Kaggle Learn rose from 30% to 50% while Coursera fell from 68% to 55%. **University courses held steady at 28%**: formal education is being supplemented, not replaced.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

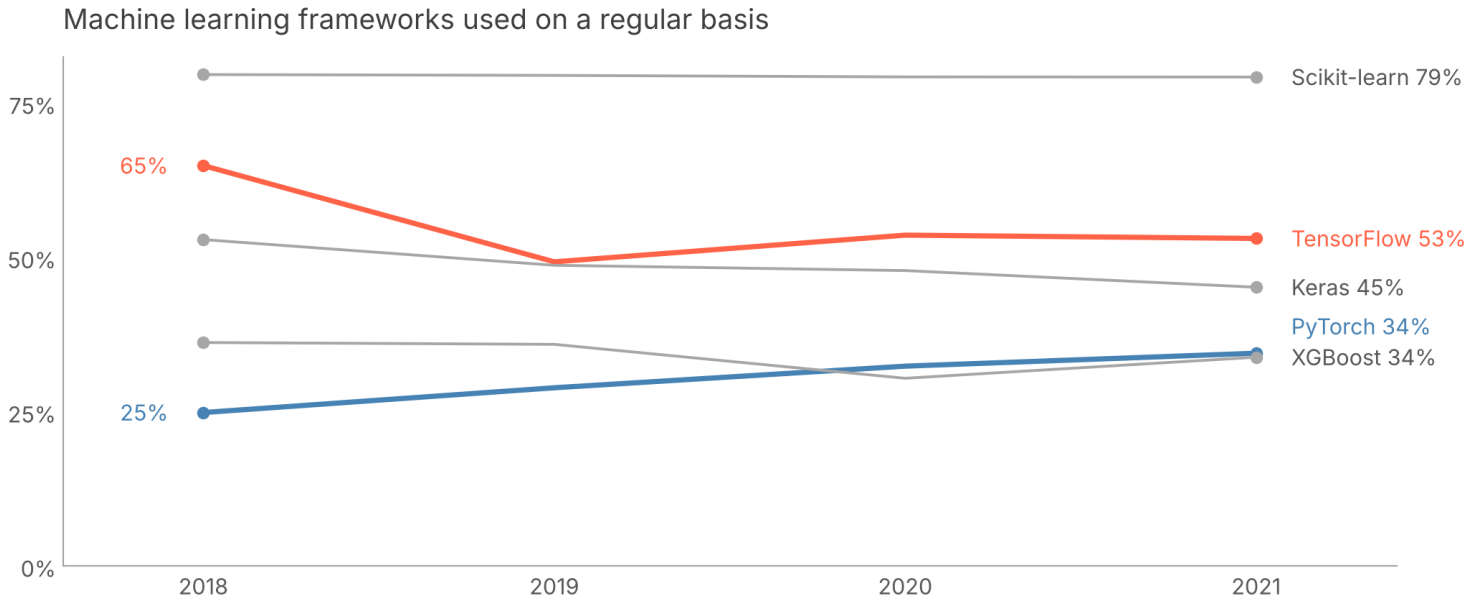
Python is the default language; R keeps losing ground while SQL holds steady



Regular R use halved from 37% to 22%, and only 6% would now recommend R to a beginner. **Python plus SQL** has become the standard skill pair.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

PyTorch is the only major framework still growing; TensorFlow and Keras are slipping

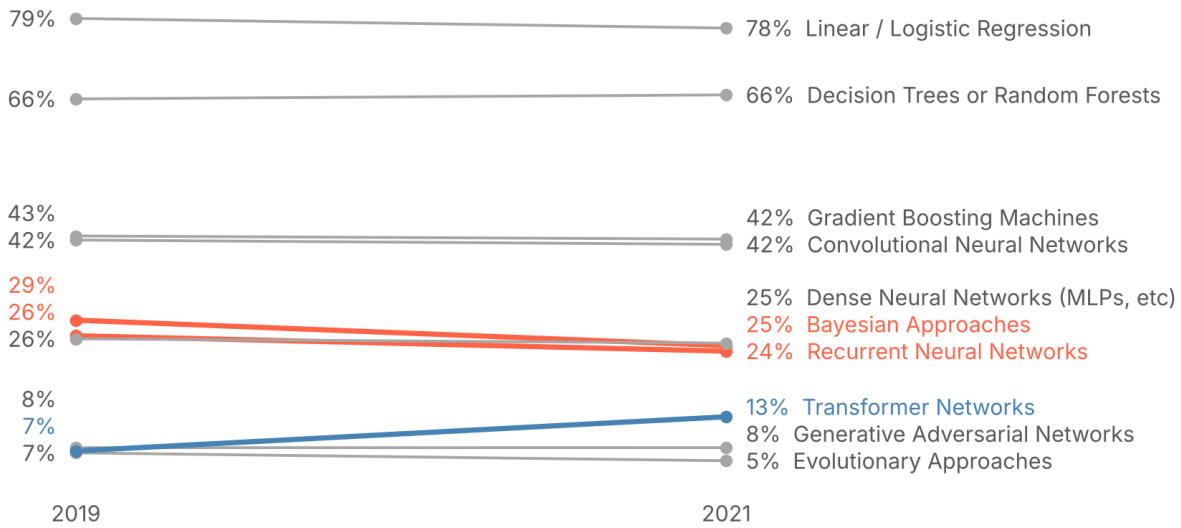


PyTorch rose from 25% to 35% while TensorFlow fell from 65% to 53%. **Scikit-learn is the constant at 79%**: most modelling is still classical.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

Everyday machine learning is still regression and trees; transformers are the only method gaining ground

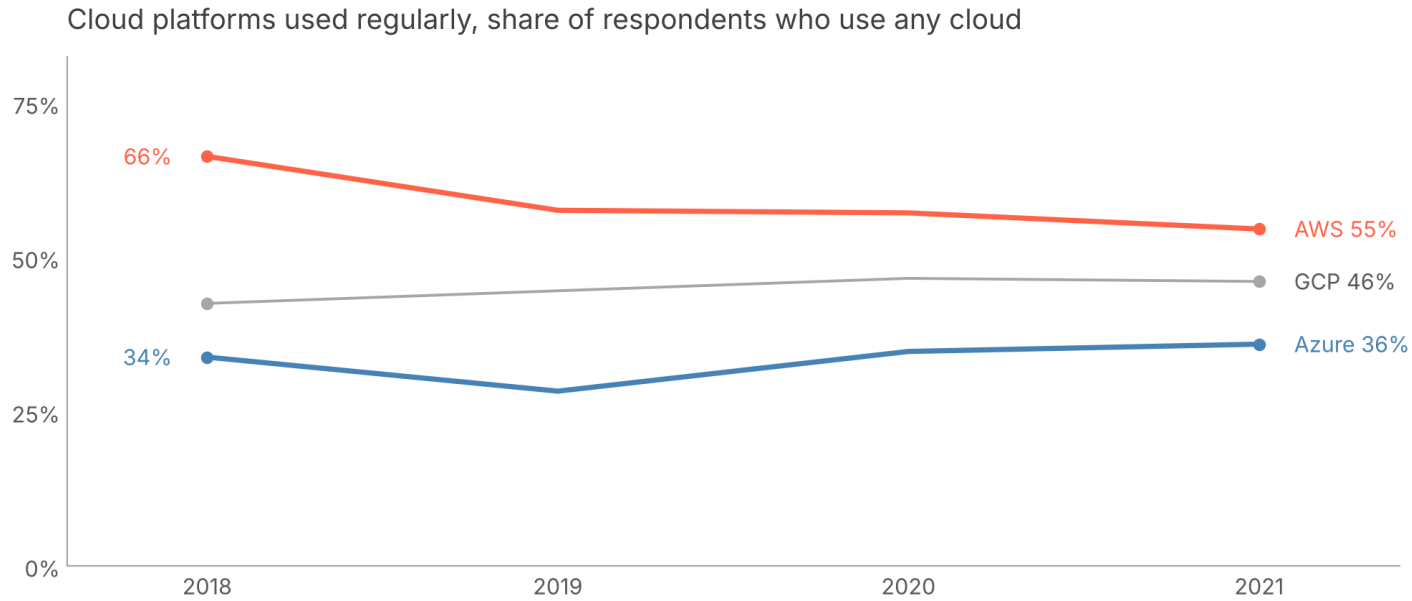
Share of respondents using each method regularly, 2019 to 2021



78% use regression and 66% use tree models, unchanged since 2019. **Transformers nearly doubled, from 7% to 13%**: the start of the shift that large language models accelerated after 2021.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

Among cloud users AWS still leads, but its lead is shrinking and Azure is closing the gap

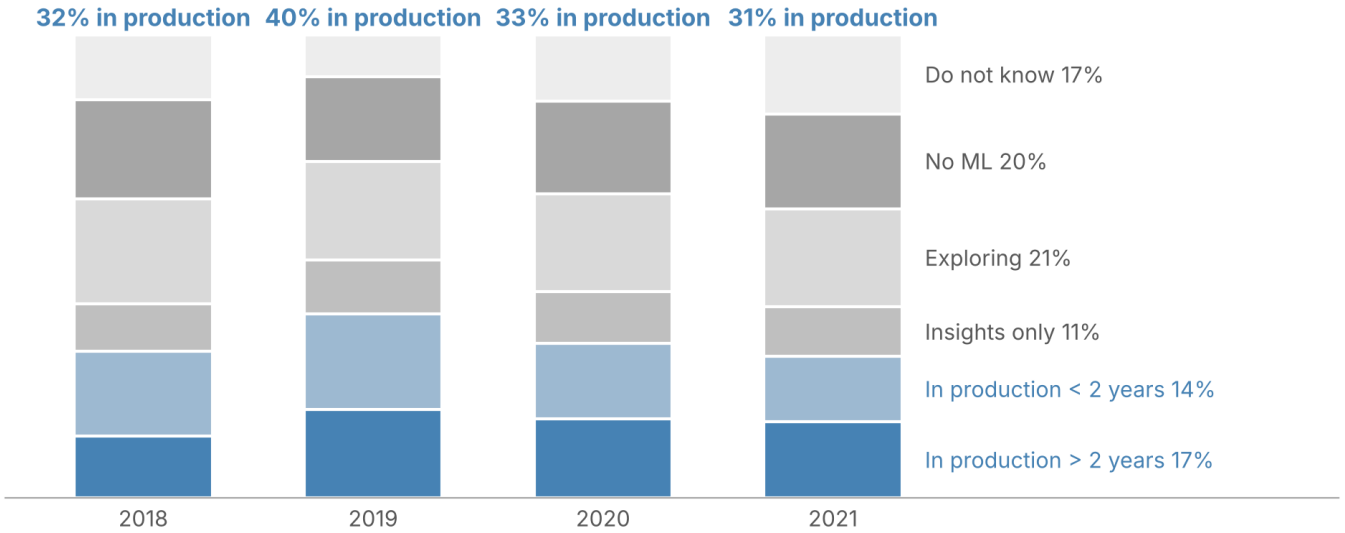


AWS fell from 66% to 55% of cloud users while Azure recovered to 36% and GCP held at 46%. The three shares sum to well over 100%: **multi-cloud is normal**.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

Only about a third of employers have machine learning in production, and that share has not grown since 2018

Employer use of machine learning, share of respondents



Skills and tools settled quickly, yet 31% of employers run models in production in 2021 against 32% in 2018, and one in five uses no ML at all. **The bottleneck is deployment, not modelling**.

Source: Kaggle Data Science Survey 2017 to 2021, 106,301 respondents. Analysis and charts: kaggle_survey_eda.ipynb

RECAP

Eight trends, one slide

WHO THEY ARE	The community grew 55% and moved from the US to India.
	It got younger and more student-heavy; women stayed under one in five.
HOW THEY LEARN	Learning moved to free, hands-on platforms; the big course providers all lost share.
HOW THEY WORK	Python plus SQL is the default; R is fading.
	PyTorch is the only major framework still growing.
	Everyday ML is still regression and trees; transformers are the only method gaining.
WHERE IT RUNS	AWS leads the cloud, its lead is shrinking, and multi-cloud is normal.
THE TWIST	**Only about a third of employers run ML in production, no more than in 2018.**

Three implications for planning hiring, tooling and delivery

1

Python plus SQL is the baseline skill set, with PyTorch for deep learning

R is now a legacy skill outside specialist niches. The talent pool is young, international and still four-fifths men, so early outreach to students and women is where the funnel can be widened.

2

Classical methods remain the centre of everyday delivery

Regression, trees and gradient boosting still carry most of the work. Transformer and LLM work is a fast-growing option rather than the default.

3

The gap to close is running models, not building them

The industry has stalled at one in three employers with ML in production. Deployment, monitoring and portable multi-cloud tooling are where investment is likely to pay off.

These are directions suggested by the survey, not measured outcomes; the notebook and appendix show how each number was derived.

APPENDIX

Method, caveats and how to reproduce every number in this deck

Method

- Source file kaggle_survey_2017_2021.csv (Kaggle, andradaolteanu). Row 1 holds the question text; 106,301 responses follow, 293 columns, 2021 question numbering.
- Single-choice questions: share of respondents per answer per year. Multi-select questions: share of respondents who answered the question and ticked that option.
- Answer labels harmonised with an alias table (gender, country, role, education, framework and platform names). Age bands grouped into four.

Caveats

- Self-selected sample of Kaggle users; Kaggle-branded options (Kaggle Learn, Kaggle Notebooks) are inflated.
- Question routing changed: cloud, employer and compensation questions reached a smaller group from 2019. Levels are comparable within a year; across years, trust direction more than magnitude.
- Languages, frameworks and cloud questions start in 2018 and algorithms in 2019, so not every trend covers all five years. The 2021 IDE columns look mis-merged in the source file, so IDE trends were left out.

Reproduce

- Run kaggle_survey_eda.ipynb top to bottom (Python 3.12, pandas, numpy, matplotlib, seaborn) with the CSV in the same folder. Every chart in this deck is written to figures/ and the headline numbers to figures/insight_summary.csv.